

1. transform Properties (Used to move, rotate, scale, skew elements)

Property	Description
transform	Shorthand to apply transformations
translateX(px / %)	Moves element horizontally
translateY(px / %)	Moves element vertically
translate(x, y)	Shorthand to move in both directions
scale(n)	Scales element (1 = 100%)
scaleX(n)	Scales horizontally
scaleY(n)	Scales vertically
rotate(deg)	Rotates the element
skewX(deg)	Skews horizontally
skewY(deg)	Skews vertically
matrix(...)	Combines all transforms in one (rarely used manually)
perspective(n)	Adds 3D perspective
rotate3d(x, y, z, deg)	3D rotation
translateZ(n)	Moves in Z direction (3D)

✅ 2. transition Properties (Used to animate changes smoothly)

Property	Description
transition	Shorthand for all below properties
transition-property	CSS property to apply transition to (e.g., all, background-color, transform)
transition-duration	Duration (e.g., 0.3s, 500ms)

Property	Description
transition-timing-function	Speed curve (ease, linear, ease-in, ease-out, ease-in-out, cubic-bezier(...))
transition-delay	Wait time before transition starts

3. animation Properties (Used for keyframe-based animations)

Property	Description
animation	Shorthand for animation properties
animation-name	Name of the @keyframes animation
animation-duration	Time it takes for one cycle
animation-delay	Delay before animation starts
animation-iteration-count	Number of times to play (infinite, 1, 2, etc.)
animation-direction	Direction of animation (normal, reverse, alternate, alternate-reverse)
animation-timing-function	Speed curve
animation-fill-mode	State before/after animation (none, forwards, backwards, both)
animation-play-state	Controls animation (running, paused)